

Comparison of Nutritional Quality of the Vegan, Vegetarian, Semi-Vegetarian, Pesco-Vegetarian and Omnivorous Diet

The number of studies comparing nutritional quality of restrictive diets is limited. Data on vegan subjects are especially lacking. It was the aim of the present study to compare the quality and the contributing components of vegan, vegetarian, semi-vegetarian, pesco-vegetarian and omnivorous diets. Dietary intake was estimated using a cross-sectional online survey with a 52-items food frequency questionnaire (FFQ). Healthy Eating Index 2010 (HEI-2010) and the Mediterranean Diet Score (MDS) were calculated as indicators for diet quality. After analysis of the diet questionnaire and the FFQ, 1475 participants were classified as vegans (n = 104), vegetarians (n = 573), semi-vegetarians (n = 498), pesco-vegetarians (n = 145), and omnivores (n = 155). The most restricted diet, i.e., the vegan diet, had the lowest total energy intake, better fat intake profile, lowest protein and highest dietary fiber intake in contrast to the omnivorous diet. Calcium intake was lowest for the vegans and below national dietary recommendations. The vegan diet received the highest index values and the omnivorous the lowest for HEI-2010 and MDS. Typical aspects of a vegan diet (high fruit and vegetable intake, low sodium intake, and low intake of saturated fat) contributed substantially to the total score, independent of the indexing system used. The score for the more prudent diets (vegetarians, semi-vegetarians and pesco-vegetarians) differed as a function of the used indexing system but they were mostly better in terms of nutrient quality than the omnivores.